

Center and Spread Data Lab



Grade 6 • Mathematics
NC.6.SP.3

Compare center and variability in data.



What will a strong explanation show?

1. A model or equation that fits the quantities.
2. Accurate calculation with appropriate units.
3. A check that makes the answer reasonable.



Watch the reasoning

Set A: 5, 7, 9, 11, 13. Set B: 5, 7, 9, 11, 23. Find the mean, median, and range of both sets. Explain what changed.

A mean = $45/5 = 9$; B mean = $55/5 = 11$. The third value is the median in each ordered set. Increasing the greatest value raises mean and range, not median.



Build a second model together

Set A: 6, 8, 10, 12, 14. Set B: 6, 8, 10, 12, 24. Find the mean, median, and range of both sets. Explain what changed.

Discuss: What should the first step find?



Your turn • 1

Set A: 2, 4, 6, 8, 10. Set B: 2, 4, 6, 8, 20. Find the mean, median, and range of both sets. Explain what changed.

Represent → solve → explain → check



Your turn • 2

Set A: 6, 8, 10, 12, 14. Set B: 6, 8, 10, 12, 24. Find the mean, median, and range of both sets. Explain what changed.

Represent → solve → explain → check



Your turn • 3

Set A: 4, 6, 8, 10, 12. Set B: 4, 6, 8, 10, 22. Find the mean, median, and range of both sets. Explain what changed.

Represent → solve → explain → check



Your turn • 4

Set A: 4, 6, 8, 10, 12. Set B: 4, 6, 8, 10, 22. Find the mean, median, and range of both sets. Explain what changed.

Represent → solve → explain → check



Exit question • Explain independently

Set A: 6, 8, 10, 12, 14. Set B: 6, 8, 10, 12, 24. Find the mean, median, and range of both sets. Explain what changed.

Include a check. What would you teach someone else?



Teacher review and next steps

1. A: 6, 6, 8. B: 8, 6, 18.

2. A: 10, 10, 8. B: 12, 10, 18.

3. A: 8, 8, 8. B: 10, 8, 18.

4. A: 8, 8, 8. B: 10, 8, 18.

Exit: A: 10, 10, 8. B: 12, 10, 18.

Use the worked PDF key and slide notes to discuss methods. Revisit the earliest unclear step before adding more practice.

